Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Stock Ticker Prediction

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Introduction

- The stock market changes continuously due to economic and social factors.
- Predicting stock prices manually is difficult because of huge and complex data.
- Machine Learning helps in analyzing historical stock data and predicting future trends.
- This project develops a stock prediction system using ML algorithms and data visualization techniques.
- The system assists investors in making better financial decisions.

Key Points:

- Data-driven prediction system
- Reduces human effort
- Helps investors in decision-making

Problem Statement

- Traditional stock analysis methods require a lot of manual effort and often fail to provide accurate predictions because of market volatility. Investors face difficulty in studying huge financial datasets and identifying useful trends.
- There is a need for an automated system that can process stock market data efficiently and generate predictions using Machine Learning algorithms. The project aims to solve this issue by developing a smart prediction model.
- Traditional prediction methods are less accurate and time-consuming.
- Investors face difficulty in analyzing market trends manually.

Objectives

Objectives of the Project

- To collect and analyze historical stock market data.
- To preprocess stock datasets for accurate prediction.
- To apply Machine Learning algorithms for predicting future stock prices.
- To visualize stock market trends using graphs and charts.
- To improve decision-making for investors through data analysis.

Technologies Used

Programming Language:

- Python

Libraries:

- Pandas for data handling
- NumPy for numerical operations
- Matplotlib for data visualization
- Scikit-learn for Machine Learning algorithms
- yfinance for collection of data of the Stock

Methodology

Step 1:
Collect historical stock market data from datasets.
Step 2:
Clean and preprocess the data by removing missing values and organizing records.
Step 3:
Train the Machine Learning model using historical stock information.
Step 4:
Generate future stock price predictions.
Step 5:
Display prediction results using graphs and charts.

Machine Learning Model

Machine Learning Algorithm
Algorithm Used:

- Linear Regression

Working:

- The Linear Regression algorithm studies the relationship between historical stock prices and future values. It learns from past trends and predicts upcoming stock prices based on patterns found in the dataset.

Advantages:

- Simple and efficient algorithm
- Fast prediction process
- Useful for trend analysis

Results & Output

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Results

- The system successfully analyzed stock market data and generated prediction results. The graphical representation helped in understanding market trends more clearly.
- The Machine Learning model produced predictions based on historical data and provided useful insights for stock analysis.

Output Includes:

- Stock prediction graphs
- Trend analysis charts
- Predicted stock values

Advantages & Future Scope

Advantages

- Reduces manual effort in stock analysis.
- Provides faster prediction results.
- Helps investors understand market trends better.
- Visual graphs improve data understanding.

Future Scope

- Integration with real-time stock market APIs.
- Use of advanced Deep Learning models for higher accuracy.
- Development of mobile and web applications.
- Addition of live market trend analysis features.

Conclusion

This project demonstrates how Machine Learning can be applied in stock market prediction. By analyzing historical stock data, the system can identify patterns and generate future stock price predictions effectively . The project also provides graphical visualization for better understanding of market behavior. In the future, advanced algorithms and real-time data integration can further improve prediction accuracy and system performance.

Thank You